

Final Project - The Analysis Report

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Introduction

This python application is a ride share application that is driven by an event engine analyses the operations of a ride share in a graph map network. It will handle riders' requests and find the best car and path for the rider's destination, and this will be done in a time loop-based frame.

It will also provide an analysis of the program that was ran with the given arguments. This would include the number of complete trips and the failed number of events along with the avg of wait time and trip duration. Will produce to images one or a summary of the rideshare and the final locations of the cars and the other will be of the timespan snapshots at the given time intervals with captures of the map.

Methodology

The simulation is built on a discrete-event engine driven by a **min-heap priority queue** that processes three main event types: RIDER_REQUEST, PICKUP_ARRIVAL, and DROPOFF_ARRIVAL. It operates within a defined time-bound loop, processing rider requests and fleet updates chronologically as events occur.

Quadtree

A 2D Quadtree spatial index is used to query and locate the nearest available vehicles on the map, optimizing candidate filtering for incoming rider requests.

Dijkstra's Algorithm

Dijkstra's algorithm calculates the shortest path and travel times between graph nodes using node coordinates and edge weights to route cars efficiently between pickup and drop-off locations.

Parameters for the Simulation Run

PARAMETER	VALUE	CLI ARGUMENT
VEHICLES	5	<code>--num-cars 5</code>
RIDERS	60	<code>--num-riders 60</code>
CANDIDATE COUNT	5	<code>--candidate-count 5</code>
MAX SIMULATION TIME	300.0 time units	<code>--max-time 300.0</code>
SNAPSHOT INTERVAL	50.0 time units	<code>--snapshot-interval 50.0</code>
ROAD MAP DATASET	Data/graph_xy.csv	<code>--graph-file Data/graph_xy.csv</code>
PSEUDORANDOM SEED	123	<code>--seed 123</code>

Results

--- METRICS SUMMARY ---

Span: 334.77 time units

Total Riders: 60

Completed Trips: 25

Unmatched/Failed: 35

Avg Wait Time: 25.72 units

Avg Trip Duration: 31.16 units

Driver Utilization: 85.0%

Console Output

=== SIMULATION STARTED ===

[0.00] RIDER_REQUEST: R1 at (25.0, 50.0) -> (75.0, 50.0)

-> Assigned Car C2 (ETA: 14.00 units)

[2.38] RIDER_REQUEST: R2 at (25.0, 75.0) -> (75.0, 50.0)

-> Assigned Car C5 (ETA: 22.00 units)

[3.25] RIDER_REQUEST: R3 at (25.0, 75.0) -> (25.0, 25.0)

-> Assigned Car C3 (ETA: 24.00 units)

[4.15] RIDER_REQUEST: R4 at (25.0, 0.0) -> (50.0, 25.0)

-> Assigned Car C4 (ETA: 0.00 units)

***** Removed *****

[273.22] PICKUP_ARRIVAL: Car C2 picked up Rider R47

[274.16] RIDER_REQUEST: R58 at (100.0, 25.0) -> (50.0, 50.0)

-> No available cars for R58.

[275.64] RIDER_REQUEST: R59 at (75.0, 0.0) -> (25.0, 75.0)

-> No available cars for R59.

[275.73] PICKUP_ARRIVAL: Car C3 picked up Rider R45

[278.16] RIDER_REQUEST: R60 at (25.0, 25.0) -> (0.0, 25.0)

-> No available cars for R60.

[285.58] PICKUP_ARRIVAL: Car C4 picked up Rider R54

[299.77] PICKUP_ARRIVAL: Car C1 picked up Rider R57

[300.58] DROPOFF_ARRIVAL: Car C4 dropped off Rider R54

[309.48] DROPOFF_ARRIVAL: Car C5 dropped off Rider R44

[312.73] DROPOFF_ARRIVAL: Car C3 dropped off Rider R45

[316.22] DROPOFF_ARRIVAL: Car C2 dropped off Rider R47

[334.77] DROPOFF_ARRIVAL: Car C1 dropped off Rider R57

===== SIMULATION COMPLETED =====

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SIMULATION CONFIGURATION SETTINGS

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Total Vehicles (--num-cars) : 5

Total Riders (--num-riders) : 60

Candidate Count (--candidate-count) : 5

Max Simulation Time (--max-time) : 300.0 time units

Snapshot Interval (--snapshot-int) : 50.0 time units

Graph Dataset (--graph-file) : Data/graph_xy.csv

Seed (--seed) : 123

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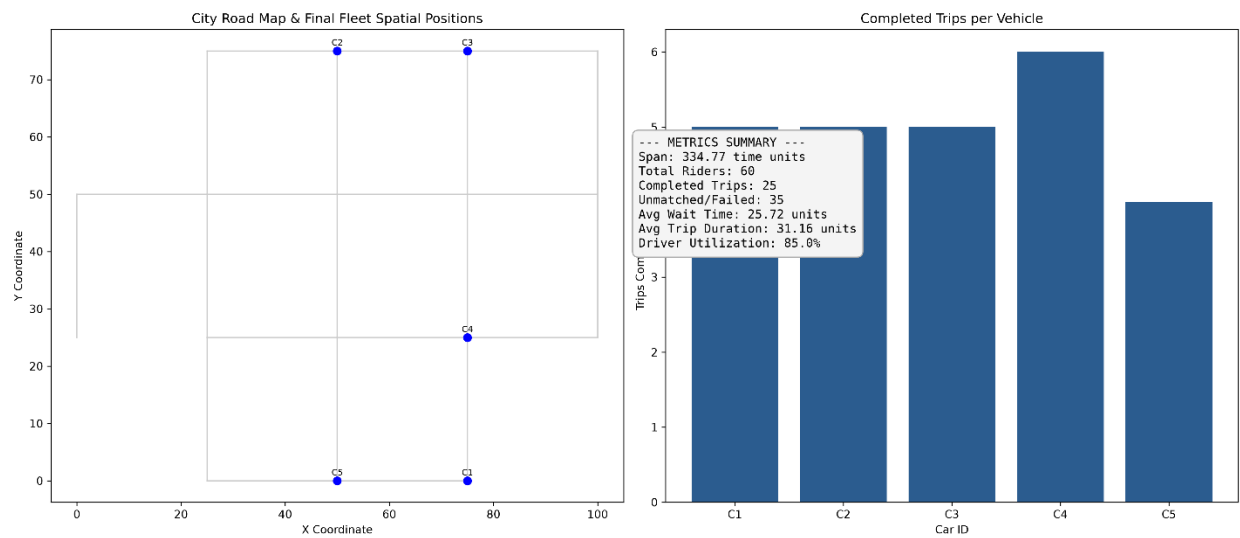
===== File Outputs =====

Saved summary chart to: simulation_summary.png

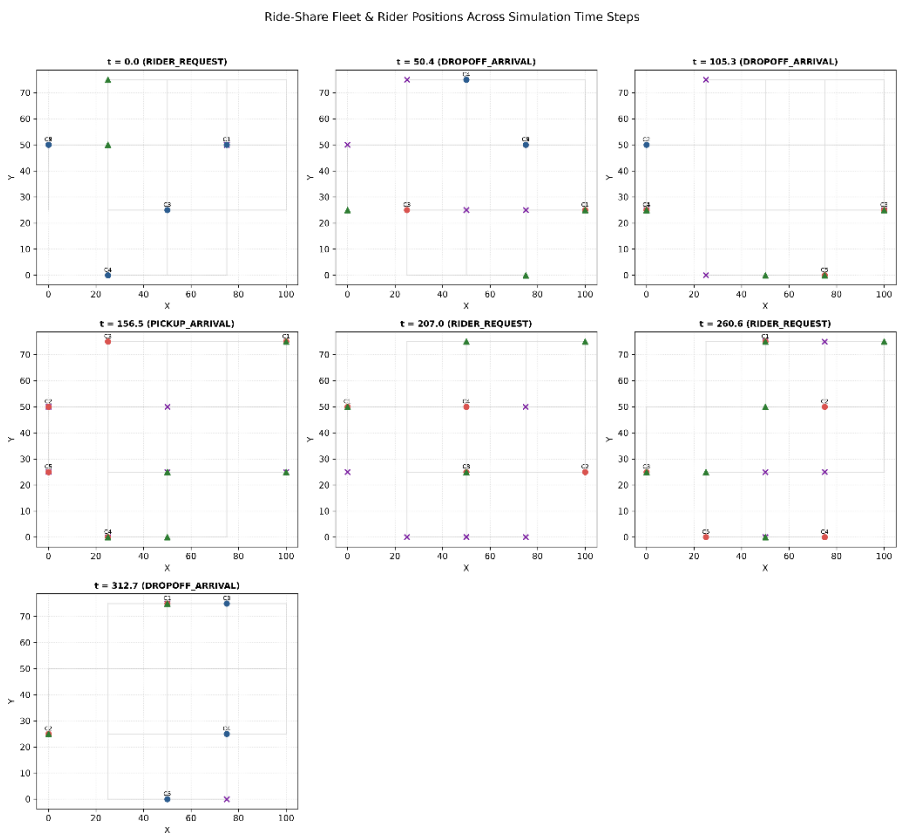
Saved time-step snapshots to: simulation_time_steps.png

Simulation completed. Check 'simulation.log' for detailed simulation logs.

Simulation Summary Image Output



Simulation Time Steps Image Output



Discussion

While running the simulation and testing different variables in the arguments of the application notice the following. The less drivers in Sim had harder driver utilization (85%) but the wait time (25.72) until a rider was picked up by a car and delivered to their location. The rider experience did start to suffer when there are not enough drivers to handle all the requests in a reasonable timeframe also would lead to riders not even being assigned a driver in the time span window of the max time of the application. For example, in the result above there were 35 unmatched/failed riders out of the 60 riders.

If the above scenario and change how many cars are in the network would allow all the riders to get a ride during the time window. With faster waiting times and less driver utilization at the end but would affect the drivers not having as many customers to service.

This application currently does not consider traffic delays such as rush hour or accidents on the road network in the graph. Another is if the same car had several passengers that are going to different locations on the map during the same trip. This could affect the results with more delays when demand or traffic causes longer wait times for the riders affecting the user experience.

Conclusion

My conclusion is that this Python application creates a basic ride share application that demonstrates the finding the closest car and best path to the rider's destination. Along with the analysis of how the network performed during the simulation run.

Future updates to this application could include Multi-destination for riders and having several riders per car. Also adding in pricing to the simulator.